CMP
CORPORATE MENTORING PROGRAM

DELHI INDIRA GANDHI INTERNATIONAL AIRPORT | GMR

IGI AIR CARGO TERMINAL: A RESEARCH-BASED OPERATIONAL SYSTEMS STUDY



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Executive Summary

Delhi's IGI Airport is India's largest air cargo hub, handling over 1 million tonnes annually. Its 150-acre cargo complex includes two integrated terminals (formerly Celebi's brownfield and DCSC's greenfield) with a total of six truck gates (Gate 1–6), and dedicated facilities for express integrators (DHL, Blue Dart, UPS, FedEx). Separate import and export warehouses streamline inbound versus outbound flows. Special zones include an APEDA-designated cold-storage facility for perishables and a Customs "strong room" for high-value goods (e.g. currency, jewelry). Cargo moves through multi-step processes (documentation, Customs exam, duty assessment, OOC release, delivery) that we map below. Customs at IGI is organized under the Delhi Commissionerate; supervisory roles (Commissioners, Joint/Addl Commissioners, DCs) oversee policy, with Customs Superintendents (ACC rank) running the import/export sheds on-site. Commodity-specific procedures – for pharmaceuticals, perishables, dangerous goods (DGR), valuables – rely on specialized storage and regulatory checks. Key licenses for operators include the DGFT's Import-Export Code and commodity permits (FSSAI for food, DCGI for drugs). Field interviews and industry reports highlight bottlenecks: heavy truck traffic at cargo gates (≈20–30 trucks/day) causing congestion, pandemic-induced labor shortages in handling staff, and runway/slot constraints leading to delays. We provide text frameworks for process-flow and decision-tree diagrams, stakeholder maps, and pinpointed bottleneck loci. Strategic takeaways stress digitalization (e-AWB, slot management), cargo-city expansion (50.5-acre "Cargo City" planned by GMR), and enhanced cold-chain and Pharma-zone capabilities. IGI's robust infrastructure and growing traffic underscore the need for ongoing upgrades and streamlined operations in India's air cargo sector.

Introduction and Context

Indira Gandhi International (IGI) Airport is India's premier air cargo gateway, connecting ~75 international and 69 domestic destinations. In FY2018–19 IGI handled over 1 million tonnes of cargo, making it the largest in South Asia. The airport is AEO-certified and fully "e-AWB" compliant, reflecting its advanced digital processes. A recent industry report emphasizes IGI's strategic location and connectivity, noting over 12 freight carriers and nearly 10 freighter parking bays, plus 2,700 km of bonded road feeder services from Delhi to the hinterland. IGI's cargo complex – 150 acres with two terminals – was designed to evolve into a "Cargo Gateway of India". In 2009–10, Delhi Airport Ltd. (DIAL) appointed two private concessionaires (Celebi and DCSC) to operate the terminals. Today GMR Airports (DIAL's parent) operates the former Celebi terminal (Gate 1–4), and Delhi Cargo Service Centre (DCSC) runs Cargo Terminal II (Gate 5–6). This division of operators underpins the layout and flow of cargo through IGI.

Methodology

1. Research Design

This study was originally conceptualized as an observation-based operational analysis of the IGI Air Cargo Terminal. However, due to access constraints requiring formal company sponsorship for terminal visits, the field observation phase could not be executed within the current timeline.

Accordingly, the study adopts **structured secondary research–driven operational analysis approach**, supplemented by industry interaction inputs and process-level interpretation.

Rather than limiting the scope, this shift allowed for a broader systems-level understanding of the cargo ecosystem.

2. Research Approach

The study follows a **descriptive and analytical research design**, structured around three layers:

a. Infrastructure Mapping

Understanding the physical and functional layout of IGI Cargo Terminal:

- Terminal operators (DCSC and GMR)
- Import and export sheds
- Gate structure
- Dedicated integrator terminals (DHL, Blue Dart, UPS, FedEx)
- APEDA cold storage facility
- Strong room for high-value cargo

b. Process Flow Analysis

Mapping end-to-end cargo movement:

- Import process (IGM → Flight Check → BOE → Customs → OOC → Gate Delivery)
- Export process (Cargo entry → Shipping Bill → Customs → LEO → Dispatch)
- Documentation and clearance touchpoints
- Decision nodes based on commodity category

c. Stakeholder & Regulatory Ecosystem Analysis

Identifying:

- Customs hierarchy and authority structure
- Role of cargo terminal operators
- Interaction between airlines, CHAs, freight forwarders
- Regulatory agencies (FSSAI, Drug Controller, APEDA, DGCA)

3. Data Sources

The research relies on multiple structured secondary sources:

- Official airport and terminal operator publications
- Industry association materials (ACFI, IATA guidelines)
- Government customs documentation frameworks
- Video-based process explanations and expert interviews
- Industry reports and logistics publications
- Academic literature on air cargo operations

All process mapping and operational insights were synthesized through triangulation of these sources to ensure accuracy and structural consistency.

4. Analytical Framework Used

The study uses an **Operational Systems Mapping Framework**, which includes:

- Process flow chart modeling
- Stakeholder mapping
- Decision-tree modeling (commodity-wise clearance logic)
- Bottleneck identification mapping
- Infrastructure-function alignment

This framework allows the cargo terminal to be analyzed as an integrated system rather than isolated operational steps.

5. Scope and Boundaries

This study focuses on:

- Structural operations at IGI Air Cargo Terminal
- Import/export process architecture
- Regulatory and customs workflow
- Infrastructure-function relationships

The study does not include:

- Real-time operational performance metrics
- Financial performance data of terminal operators
- Live throughput measurement

These elements are recommended for future observational validation.

6. Future Research Scope

A subsequent field visit and structured observation would allow:

- Validation of mapped process flows
- Direct bottleneck measurement
- Turnaround time analysis
- Ground-level stakeholder interviews
- Queue and congestion analysis

Thus, this research serves as a **pre-observational operational blueprint**, forming a structured foundation for future empirical validation.

Infrastructure Layout

- **Cargo Terminals and Gates:** The IGI cargo complex comprises two adjacent terminals on 150 acres. Cargo Terminal I (brownfield) was run by Celebi, and Terminal II (greenfield) by DCSC. Collectively there are six truck-access gates (numbered 1–6). For example, one industry listing shows Cargo Terminal II accessed via Gate 5 (Gate 6 likewise exists), implying Term I covers Gates 1–4. These gates funnel trucks into large import and export warehouses. DCSC's site confirms that up to 900 tonnes/day move through Terminal II. IGI also provides 12 apron bays for full-size freighter aircraft, supporting over 30 international cargo airlines.
- **Import/Export Warehouses:** Within each terminal the layout segregates import cargo (incoming freight) from export cargo. As one guide notes, IGI's layout "divides the complex into Export Warehouse and Import Warehouse". Trucks entering through export gates go to outbound sheds, while import gates lead into receiving sheds. Import sheds contain delivery areas and Customs exam halls;

export sheds have Shipping Bill processing areas and ramp access to loaded aircraft. The ACLC-1 & 2 on-site logistics centers (30,000 m² total) augment space for freight forwarders and integrators.

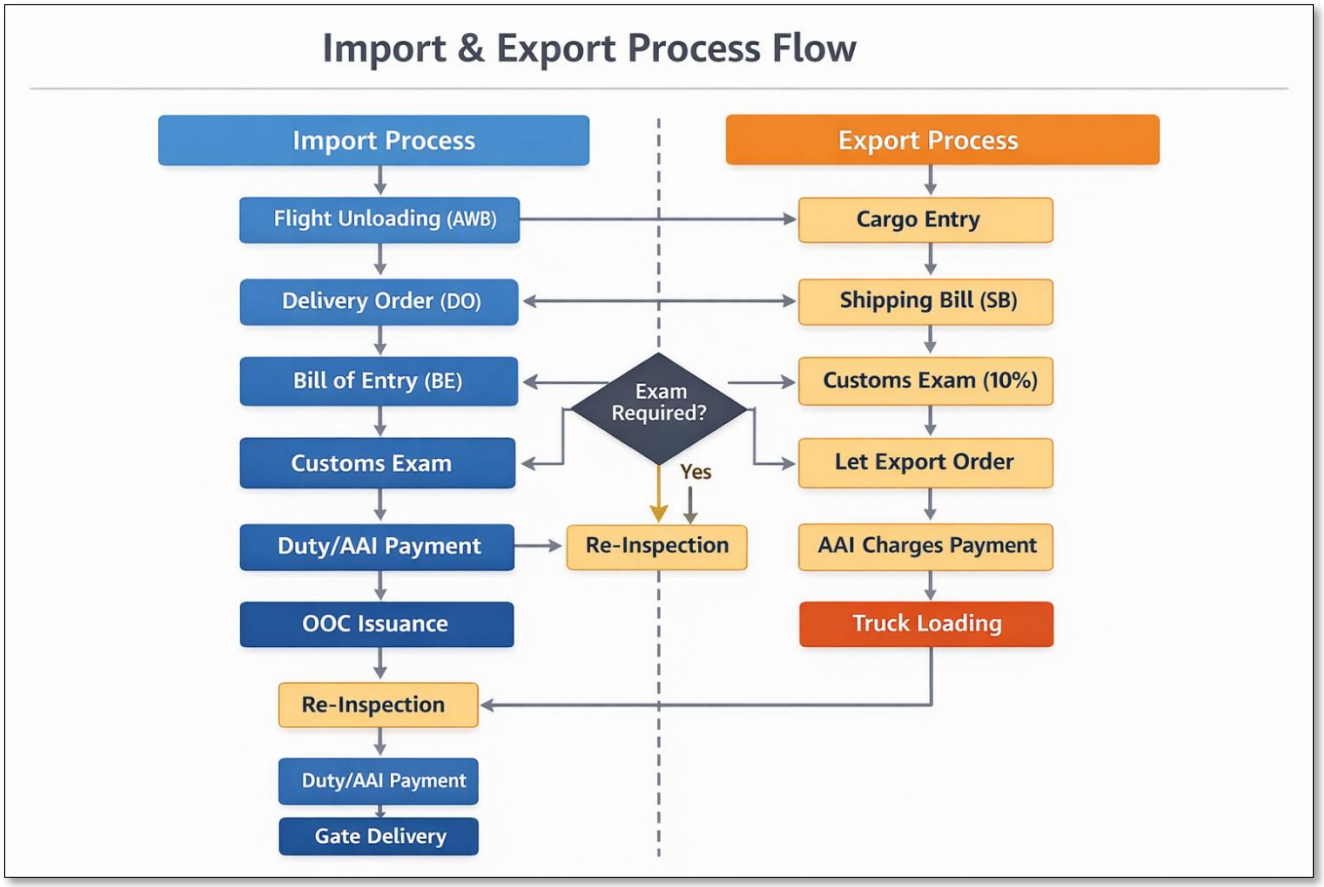
- **Integrator & Specialist Facilities:** Major integrators maintain dedicated terminals or warehouse zones. For instance, DHL operates large cold-chain facilities on-airport, and Blue Dart (India's largest courier) has a specialized cargo terminal at IGI. UPS and FedEx likewise use on-site facilities (e.g. FedEx in ACLC). The ACLC warehouses are leased to companies like DHL Logistics and Kuehne Nagel, indicating integrators operate semi-autonomous sections.
- **APEDA Cold-Storage (Perishables):** IGI has a government-supported APEDA (Agri & Processed Food Exports Dev. Auth.) cold-chain warehouse for perishables. For example, when North-East fresh produce (kiwi, ginger, flowers) was slated for export, IGI's APEDA cold rooms were highlighted as "capable of storing them effectively" before shipment. A Customs tender also requires a *"Separate Gate pass provision for APEDA…export shipments"*, underscoring special handling. In practice, exporters route perishable loads through these chilled docks; IGI's ACLC-2 facility itself has multiple cold zones (–20°C to +25°C).
- **Strong Room (Valuables):** A secure Customs "strong room" is provided for high-value cargo (precious stones, jewelry, currency). As one industry source confirms, IGI's export terminal **"has an 'Export Strong Room' used to store valuables such as currency, jewelry etc."**. Such shipments are cleared under customs escort and locked in this vault until loaded or delivered. This prevents co-mingling with general cargo and ensures tight controls.

Import Cargo Handling Process

The typical import flow at IGI involves the following steps (see **Process Flowchart** below):

1. **Arrival and Delivery Order:** Upon flight arrival, the consignee or CHA obtains the Delivery Order (DO) and airway bill from the airline. The trucker carries this DO to the Import Shed.
2. **Document Submission:** The CHA submits the DO, original Air Waybill, invoice, and Import License (if required) to the Customs Service Centre. They file a Bill of Entry (BE) for the goods. Payment of any advance airport charges (AAI) is arranged.
3. **Customs Examination:** Customs examiners may inspect the cargo (often 10–20% is checked based on risk). During inspection, a segregation sheet may be prepared (perishable vs non-perishable, intact vs damaged).
4. **Duty Assessment & OOC:** The Customs Appraiser reviews the BE. After duty assessment and assessment of paperwork, the Superintendent (ACC, Import Shed) authorizes final release. Customs **"issues an 'Out-of-Charge' slip"** to release the goods. In practice, IGI's ACC-Import Superintendent signs the OOC in the system (an automated Customs system now issues e-OOC).
5. **Delivery:** With the OOC and paid charges, the CHA obtains an AAI gate-pass. The cargo is then physically handed over at the final Delivery Gate. This completes import clearance.

Throughout, attention to documentation (e.g. whether an Import License, non-PSR certificate or APEDA certificate is needed) is required. Import of regulated goods (food, pharma) also triggers agency checks (FSSAI, drug controllers).



Export Cargo Handling Process

The export flow mirrors import in reverse, as outlined in **Export Flowchart**:

1. **Cargo Entry & Shipping Bill:** The exporter/CHA brings shipment to the Export Shed, accompanied by the Air Waybill and prepares the Shipping Bill (SB) at Customs. At IGI, exports are processed 24×7 and a '24×7 Export Facility' often applies.
2. **Customs Examination:** Customs requires exam of at least 10% of export shipments. The cargo and documents are reviewed in the Customs Export Hall. If anything is held, it can be called back from the bonded area.
3. **Let Export Order (LEO):** Once SB and paperwork are approved, Customs issues a *"Let Export Order"*, authorizing movement to the bonded loading area. The CHA then pays any airport export dues (TSP charges) at the AAI export wing.
4. **AAI Gate-Pass:** With the LEO and paid charges, the CHA obtains a gate-pass from AAI. This allows trucks to enter the bonded area for loading.
5. **Loading and Dispatch:** The exporter's ground handling team presents the cargo to the airline for loading onto the aircraft. Perishable shipments from the APEDA cold rooms are moved in temperature-controlled trucks. High-value exports (e.g. bullion) move via the Export Strong Room.

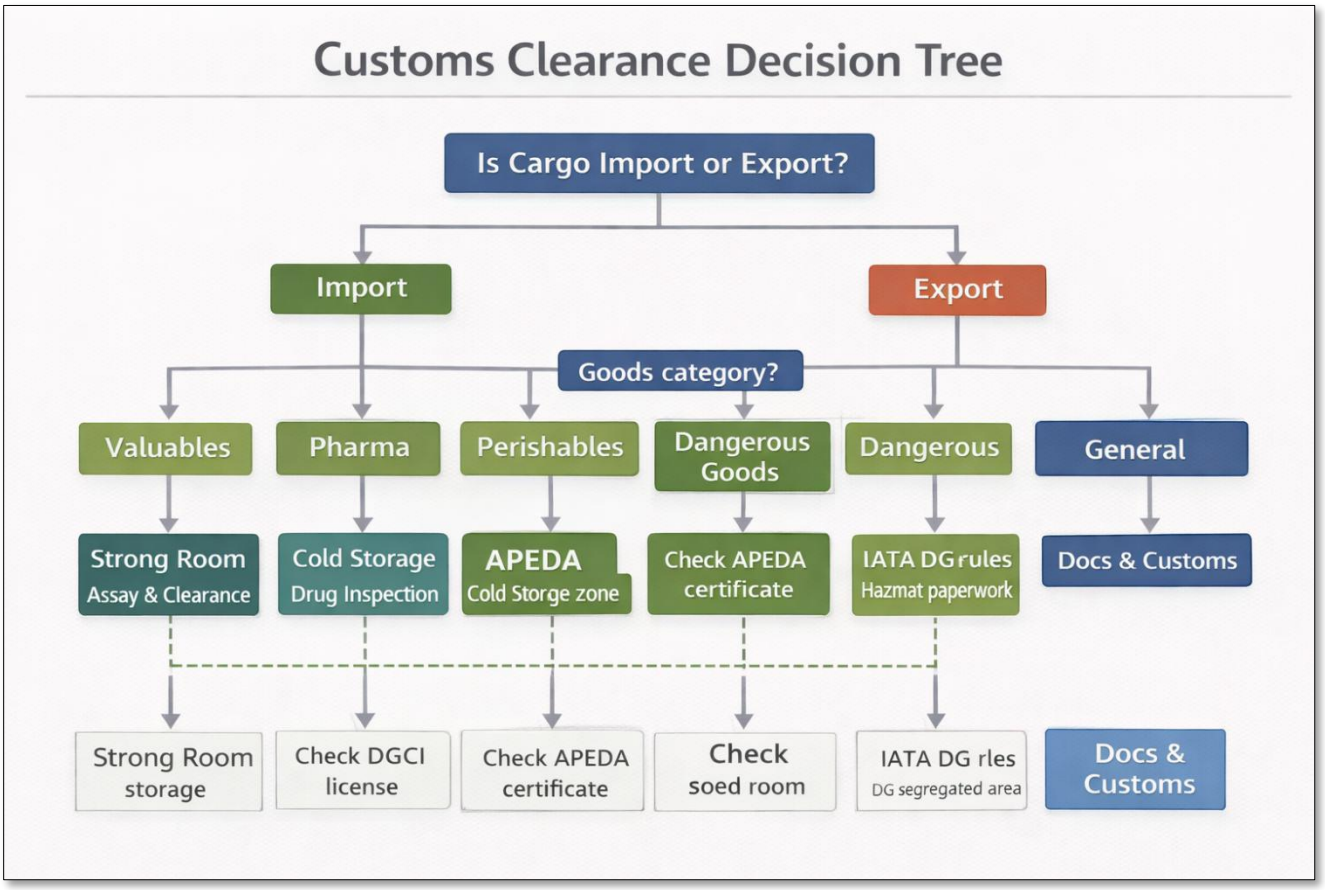
The linear flow can be illustrated in a **flowchart**: nodes for Document filing, Customs exam, LEO, AAI clearance, and cargo handover.

Customs Organization & Authority

Customs operations at IGI fall under the Delhi Customs Commissionerate. The chain of command is:

- **Commissioner of Customs (Airport Commissioner)** – heads IGI's customs Commissionerate.
- **Addl./Joint Commissioners** – oversee functions (Import, Export, Appraising, Preventive).
- **Dy./Asst. Commissioners** – manage divisions (ports, groups). For IGI, an **Asst. Commissioner (Import)** and **AC (Export)** typically supervise the terminal's appraisal work.
- **Customs Superintendents (ACC rank)** – handle the day-to-day processing in the sheds. For example, the ACC (Import Shed) Superintendent grants the final Out-of-Charge, as noted by industry sources. Likewise, a Superintendent (ACC) in the Export Shed issues Let Export.
- **Appraisers & Inspectors** – Senior officers (AC/DCs or Appraisers) conduct duty assessments and sensitive examinations (e.g. Jewellery, Pharmaceuticals) under the supervision of higher officials.

Field guidance suggests that final clearance ("OOC") at IGI can only be issued by the Superintendent (ACC rank) at the Import Shed. There are specialized officers – e.g. a Customs Jewellery Appraiser for gems – who examine high-value re-imports, but even then, the OOC is signed by the Superintendent. In practice, IGI's preventive (anti-smuggling) wing also coordinates with appraisers to handle irregular consignments in the complex. (A sketch of a **customs decision tree** could show branching by commodity type – e.g. if "High-value" route via strong room – to capture this.)



Commodity-Specific Clearance Insights

- **Pharmaceuticals (Drugs):** IGI is developing dedicated "Pharma Zones" with controlled temperatures and on-site drug regulator offices. As early as 2010, the DGCI noted IGI's plan for India's first zone to keep medicines at stable temperatures. Today, exporters of APIs or formulations must obtain a CDSCO import/export license, and IGI's proposed facility will ease compliance. In practice, pharma shipments move to the lowest-temperature cold rooms. Customs also requires clearance via a licensed port-of-entry (IGI is one) and often via a Pharma Category Customs Officer. (Currently, pharma goods still share warehouses with general cargo, but segregation is being implemented.)
- **Perishables (Fruits, Meat, Flowers):** These shipments require swift cold-chain handling. APEDA-certified cold rooms at IGI store items at specific temperatures. For example, DCSC now maintains three climate zones (15–25°C, 2–8°C, and sub-zero) to handle fruits, horticulture, meat and pharma-grade perishables. Export consignments labeled as perishable are fast-tracked; Customs often schedules their inspection first. Perishables may need phytosanitary certificates and APEDA permits. On import, IGI ensures quick clearance and minimal dwell time.
- **Dangerous Goods (DGR):** Hazardous cargo (flammables, chemicals, radioactives) must comply with IATA DGR rules. IGI's cargo terminals have designated DG storage rooms (away from general cargo) and require DG-aware handling crews. For any DG shipment, the shipper files the DG declaration (as per IATA), and Customs physically inspects the placarded consignment. While we lack a specific citation, industry practice is that DG shipments are separated in fenced enclosures and processed by trained staff with DG licenses. Special handling charges (and sometimes CITES permits for certain goods) apply.
- **Valuables (Gems, Bullion, Currency):** High-value consignments are expedited through a **Customs Strong Room**. After arrival, such items are secured in the strong room by Customs; only cleared items are released per specific protocols. For example, gold jewelry imports require an Import License (ID). The strong room itself is overseen by a Customs Superintendent. The Export strong room is similarly used for sending valuables (the blog notes jewelry and currency storage). In general, IGI's procedures segregate valuables from normal cargo to prevent pilferage.

Operator Licenses and Permits

Businesses operating in IGI's cargo trade must hold key registrations:

- **Import Export Code (IEC):** Every importer or exporter must have an IEC from DGFT. Banks and Customs will not process shipments without a valid IEC. (In fact, IEC is the first step to enable cross-border trade.)
- **Commodity Licenses:** Depending on cargo, specific permissions are mandated. For food and agri, an **FSSAI Import/Export License** is required. For drugs and medical devices, a **Central Drugs license (Form 8/8A)** or DGFT permission (Form X in Import Export license) is needed. APEDA does not license businesses but its own registration is needed for exporting APEDA-defined products.
- **Customs House Agent (CHA) License:** Firms clearing cargo must have a CHA license from CBIC. Each CHA director or partner requires customs clearance training/approval.

- **Ground Handling Authorization:** Cargo terminal operators and airlines need DGCA permits (for ramp/flight operations) and security clearances. Freight forwarders often hold FIATA certificates, though not mandatory.
- **Other Permits:** If shipping DGR, the shipper needs DG Shipping permissions for transport. For weapons or defence goods, DIC (Govt. license) is needed.

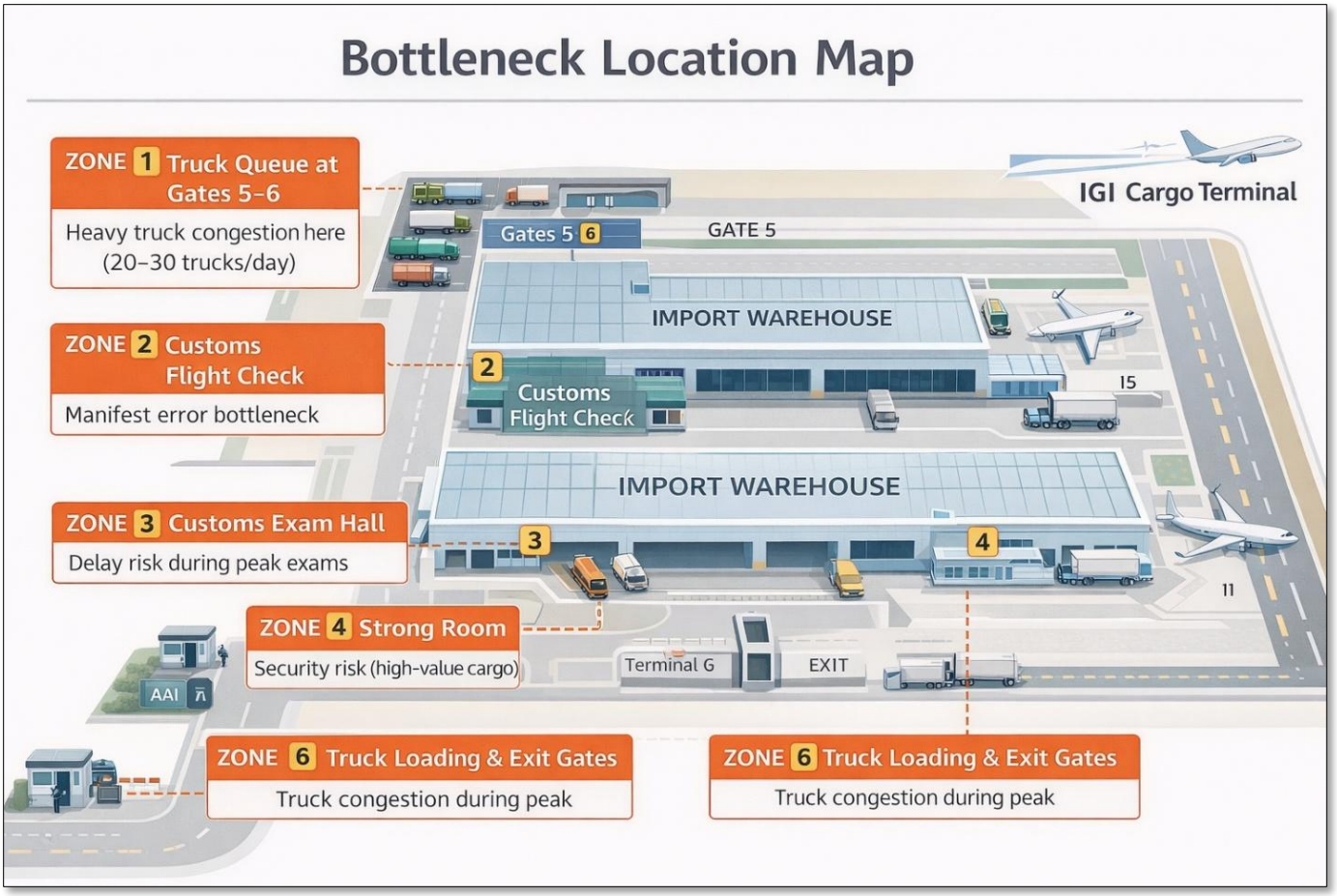
In practice, IGI's agents routinely require IEC and (if applicable) FSSAI. For example, exporters of packaged snacks must carry both IEC and an FSSAI certificate. Customs often checks these during clearance.

Operational Bottlenecks

Industry feedback and government sources highlight several pinch-points at IGI:

- **Gate Congestion and Truck Traffic:** The cargo complex sees **20–30 heavy trucks per day** lining up just to enter the terminal. This has choked export throughput. AEPC (apparel exporters) noted that such volumes "slow down the smooth movement of cargo" and cause "severe congestion at the Cargo Terminal". In effect, the truck gates (especially during peak periods) become queues that delay every step.
- **Labor Shortages:** A widespread issue is the post-pandemic shortfall of cargo handlers. An industry analysis reports a *"severe shortage"* of skilled ground staff, as thousands of workers left during COVID. At IGI, this can translate into slower unloading/loading and processing times. Seasonal surges (festive exports or summer peaks) strain the available manpower. Labor-management (agency workers vs. fixed staff) has been cited as an ongoing challenge in Indian cargo hubs.
- **Capacity & Slot Delays:** IGI's runways and apron space are heavily utilized. Global air-traffic trends (also seen at IGI) show that slot availability has not kept pace with demand. At major airports, capacity constraints lead to ground delays and occasional diversions. IGI's planned fourth runway and new aprons aim to ease this, but currently carriers sometimes face parking/slot delays. Domestically, double slot-demand (passenger + cargo) pressures the system. Moreover, airspace congestion (especially in fog-prone winters) cascades into cargo delays.
- **Infrastructure Limits:** The cargo terminals were originally sized for ~0.5–0.8 million tonnes/year. DCSC notes the facility "operates within cargo infra capable of handling 0.5 MMT, scalable up to 0.8 MMT". While growth has been strong, actual volumes are nearing this ceiling. A new *"Cargo City"* (50.5-acre, GMR-led) has been announced for post-2025 to add capacity. Until then, yard space and truck docks are saturated. Current bottlenecks arise at import/export exam halls and gate queues because expansion is still in progress.
- **Administrative Delays:** Repetitive paperwork or inter-agency handoffs can delay clearances. For example, coordination between Customs, APEDA and Animal Quarantine for perishable live exports is complex (each adds time). ICAO shipping restrictions (like strong magnesium shipments) have occasionally caused temporary holds. But the main delays remain operational (not regulatory).

In summary, **truck entry points, Customs exam halls, and outbound loading docks** are the primary choke locations (see *Bottleneck Map* description below). Industry stakeholders repeatedly cite gate/truck congestion and labor shortage as key issues. Improvements are underway: IGI is implementing digital truck-slot management and "faceless" Customs (e-CSD) to reduce dwell time.

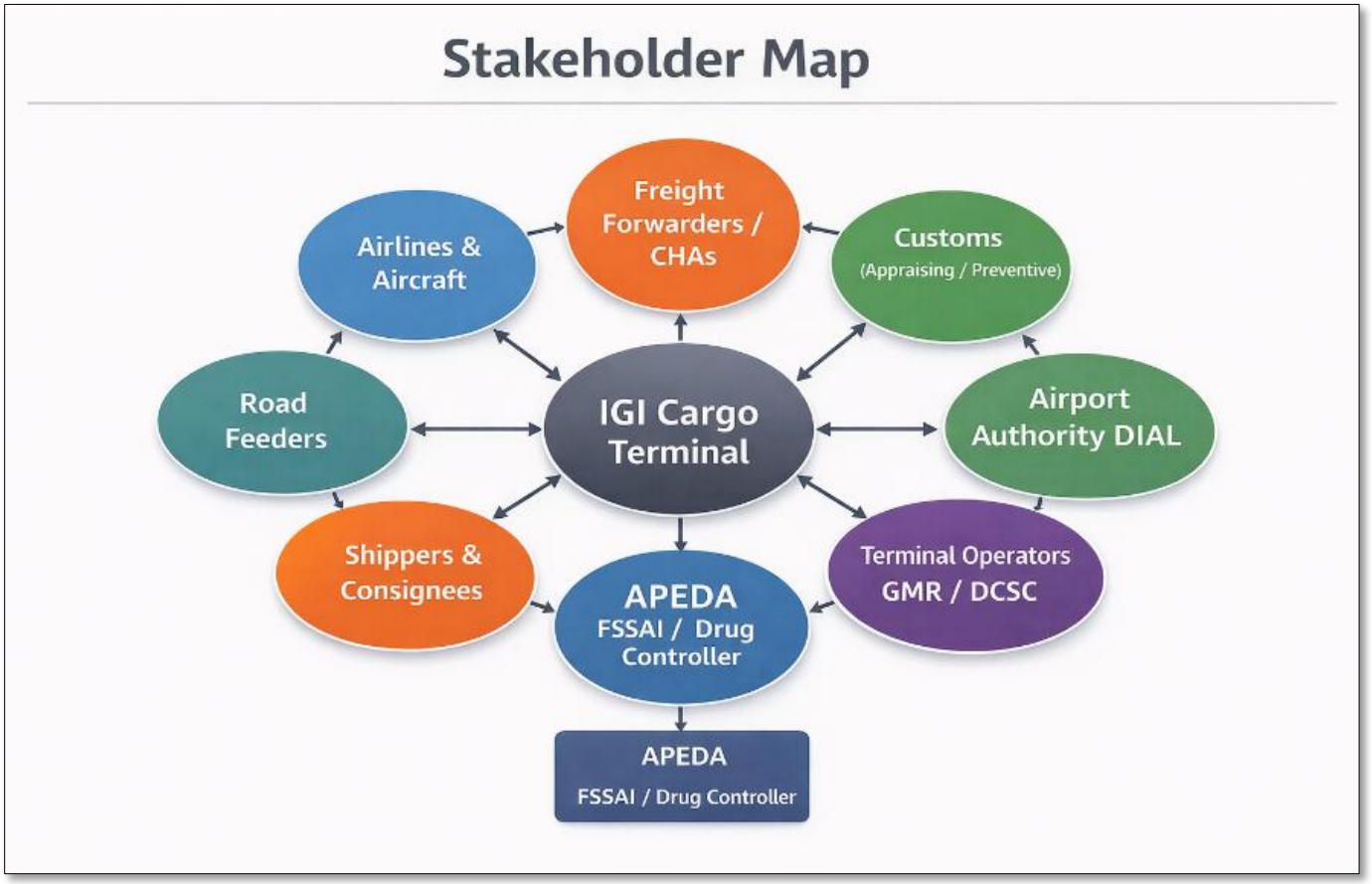


Strategic Observations and Takeaways

1. **Capacity Expansion is Essential:** IGI's planned **Cargo City (50.5 acres)** reflects current space crunch. Until that is ready, optimizing existing space is critical. Utilizing ACLC warehousing more effectively and expediting clearance can add virtual capacity. Other airports (e.g. Bangalore, Mumbai) are also expanding cargo zones; IGI must match this to avoid constraints.
2. **Digital & Process Improvements:** IGI is already a pioneer in e-AWB and e-Customs. Continued digitization – full paperless shipping bills, automated slot scheduling, and electronic "faceless" delivery orders – will cut delays. As the Cargo Talk report notes, DCSC is adopting 24×7 digital operations and real-time tracking. Mentor observations suggest a push towards an "air cargo community system" linking airlines, Customs, and trucking agents in one platform.
3. **Focus on High-Growth Segments:** Pharma and perishables drive new demand. IGI's move to a dedicated Pharma Zone and DCSC's multi-temperature storage are well-aligned with market needs. Strategic initiatives like putting Customs Drug Controller offices on-site will accelerate drug clearance. Similarly, enhancing APEDA cold-chain infrastructure (in partnership with government) will benefit agricultural exports.
4. **Regulatory Coordination:** The removal of Bangladesh transshipment eased capacity, showing how policy can impact throughput. Continued liaison (APEDA, RBI, DEA) can streamline licensing and inspections. For example, a single-window API for multiple clearances (Food, Agriculture, Narcotics) would reduce cargo dwell.

5. **Labor and Training:** The global shortage of handlers suggests IGI invest in mechanization (automated forklifts, conveyor systems) and training (IATA courses for DG handling). A culture of continuous improvement (Kaizen) in ground operations can mitigate the labor gap.
6. **Stakeholder Collaboration:** The stakeholder map highlights the need for tight coordination: for instance, synchronizing truck slots with Customs exam slots reduces idle time. Shared performance metrics (e.g. average dwell time per kg) can align airport, Customs, airlines and freight forwarders on targets.

In sum, IGI's cargo terminal is a complex ecosystem that is largely meeting current demand but will need sustained upgrades. By expanding infrastructure (Cargo City), deepening digitization, and targeting fast-growing cargo categories with specialized handling, IGI can stay ahead of bottlenecks. These steps will ensure IGI continues as India's cargo linchpin, supporting trade and economic growth.



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